Department of Electrical and Electronics Engineering
Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. ECE
Course Code & Title: CME365 Renewable Energy Technologies
Name of the Faculty member (s): Mr. S. Meenakshi Sundaravel

Innovative Practice Description

- **Unit / Topic: Unit I** / Types of energy sources and comparison between conventional and non-conventional sources
- **Course Outcome:** CO1
- **Unit Outcome:** TLO1
- **Activity Chosen:** One Minute Paper
- **Justification:**
  The types of energy sources and differences between the conventional and non-conventional sources are identified using this approach. A one-minute paper is used for this concept because it prompts students to quickly summarize key points and reflect on their understanding of existing energy resources. This technique helps in analyze the energy sources and improves the students' learning with the focused thinking.
- **Time Allotted for the Activity:** 5 minutes
- **Details of the Implementation:**
  At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement in the particular class.
- **CO – PO / PSO mapping:**

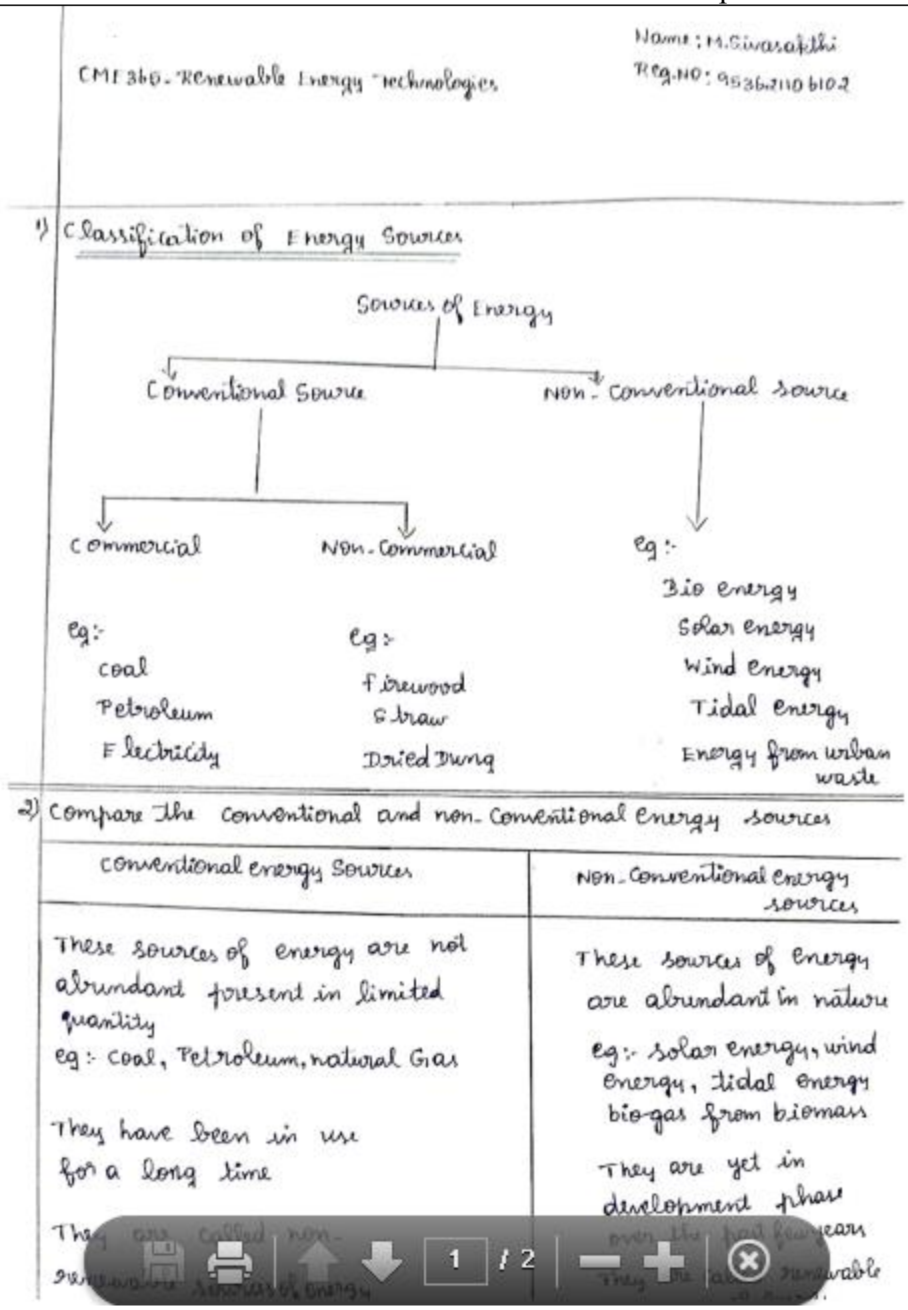
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	1	1	1	-	1	-	1	-	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 10**
	1
Justification for correlation	The students can effectively communicate on engineering activities and make presentations on the given topics

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Types of energy sources and comparison between conventional and non-conventional sources – One Minute Paper


CME 360 - Renewable Energy Technologies

Name: M.Sivasakthi
Reg.No: 953621106102

1) Classification of Energy Sources

2) Compare the conventional and non-conventional energy sources

Conventional energy Sources	Non-Conventional energy sources
These sources of energy are not abrundant present in limited quantity eg:- Coal, Petroleum, natural Gas	These sources of energy are abrundant in nature eg:- solar energy, wind energy, tidal energy bio-gas from biomass
They have been in use for a long time	They are yet in development phase over the past few years
They are called non-renewable sources of energy	They are called renewable

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ***Benefit of the practice:***
- Students can be able to attend the question even the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

1. Fundamentals and Applications of Renewable Energy | Indian Edition, by Mehmet Kanoglu, Yunus A. Cengel, John M. Cimbala, cGraw Hill; First edition (10 December 2020), ISBN-10:9390385636.
2. Renewable Energy Sources and Emerging Technologies, by Kothari, Prentice Hall India Learning Private Limited; 2nd edition (1 January 2011), ISBN-10:8120344707.

Signature of Faculty Member **HOD**

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Innovative Practice Description

Unit / Topic: Unit II / Demonstration of Solar thermal collector and PV System Operation

- **Course Outcome:** CO 2
- **Unit Outcome:** TLO11
- **Activity Chosen:** Field Visit
- **Justification:**
  The demonstration of the types of collectors and various applications of solar PV systems are virtually shown to the students through this type of approach. The power rating of the installed solar panels and working of solar water heating systems are explained. This technique helps in understanding and analyzing the various applications of the PV systems virtually and hence improves the students' learning.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**
  Field visit require students to actively engage in their learning, often by connecting their prior knowledge to new information. When going for such a visit, a student will be understanding the basics and identify the specific topics in solar thermal applications. The students are actively participated in this event.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	1	1	1	-	1	-	1	-	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 1**
	3
Justification for correlation	The students will have strong fundamental knowledge.

- **Images / Screenshot of the practice:**

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  Students understood the concept which was reflected from their answers for the questions I have asked during field visit.

- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

  The benefits of field visits are a great way for students to make notes on all of the information they receive. It made the students to identify the basic concepts in construction and working principle of solar PV systems involved in the process and make the students to deliver the same for the particular topic.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

1. Fundamentals and Applications of Renewable Energy | Indian Edition, by Mehmet Kanoglu, Yunus A. Cengel, John M. Cimbala, cGraw Hill; First edition (10 December 2020), ISBN-10:9390385636.
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Innovative Practice Description

- **Unit / Topic: Unit V** / Environmental impact of ocean and geothermal energy
- **Course Outcome:** CO 5
- **Unit Outcome:** TLO27
- **Activity Chosen:** One Minute Paper
- **Justification:**
  The environmental impact of other energy sources is identified using this approach. A one-minute paper is used for this concept because it prompts students to quickly summarize key points and reflect on their understanding of other energy resources apart from solar and wind energy. This technique helps to analyze the environmental sustainability and its effects with the usage of ocean and geothermal energy sources.
- **Time Allotted for the Activity:** 5 minutes
- **Details of the Implementation:**
  At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement in the particular class.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	1	1	1	-	1	-	1	-	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 10**
	1
Justification for correlation	The students can effectively communicate on engineering activities and make presentations on the given topics.

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Environmental impact of ocean and geothermal energy – One Minute Paper
CME 365 - Renewable Energy Technologies Environmental impact of ocean and Geothermal energy. S. Suvalakshmi 953621106112 ECE => The geothermal fluids have dissolved solids entrained solid particles, non condensable gases and sand. => These solids & non condensable gases create following Environmental problems. Emissions: => The emissions of H_2S, CO_2 and other toxic gases into the atmosphere during operation of the plant is a source of pollution to the environment. => H_2S gas has annoying smell geothermal plant may emit about 520kg of H_2S gas per day in the atmosphere. Effluents: => The effluents from geothermal fluids during conversion are highly mineralised. => These effluents can pollute the water streams or underground water source if disposed of without treatment. => Certain effluents contains borons, fluorines, mercury compounds which are extremely harmful to animals & plant life. 2 / 2

Reflective Critique:

- ❖ ***Feedback of practice from students and other stakeholders:***
- ❖ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ***Benefit of the practice:***
- Students can be able to attend the question even the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

1. Fundamentals and Applications of Renewable Energy | Indian Edition, by Mehmet Kanoglu, Yunus A. Cengel, John M. Cimbala, cGraw Hill; First edition (10 December 2020), ISBN-10:9390385636.
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